

EE 210L: Digital Circuits and Logic Design Lab

Lab 2: Boolean Simplification

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels only, 0 or 1. Enter your pre-lab predictions before you measure, and leave them as written even if a measurement disagrees.

Part 5.1: Power Supply Setup

Measured power supply voltage: _____ V

Current limit setting: _____ mA

Part 5.2: Circuit (a) — Original

#	A	B	C	D	E	Predicted F	Measured F
1	0	0	0	0	0		
2	1	0	0	0	0		
3	0	0	0	0	1		
4	0	1	1	1	0		
5	0	1	1	0	0		

Did every row match your prediction? If not, what was wrong? _____

Part 5.3: Circuit (a) — Simplified

Simplified expression $F =$ _____

Draw your simplified circuit in the box. Label every chip and pin number you used.

Figure 1: Simplified form of circuit (a).

#	A	B	C	D	E	Measured F	Same as 5.2?
1	0	0	0	0	0		
2	1	0	0	0	0		
3	0	0	0	0	1		
4	0	1	1	1	0		
5	0	1	1	0	0		

Part 5.4: Circuit (b) — Original

#	A	B	Predicted Y	Measured Y
1	0	0		
2	0	1		
3	1	0		
4	1	1		

Did every row match your prediction? If not, what was wrong? _____

Part 5.5: Circuit (b) — Simplified

Simplified expression $Y =$ _____

Draw your simplified circuit in the box. Label every chip and pin number you used.

Figure 2: Simplified form of circuit (b).

#	A	B	Measured Y	Same as 5.4?
1	0	0		
2	0	1		
3	1	0		
4	1	1		

Part 5.6: Gate and Chip Count

Circuit	AND gates	OR gates	NOT gates	Total gates	Packages
(a) original					
(a) simplified					
(b) original					
(b) simplified					

How many gates does the simplified circuit (b) need, against the seven in the original? What does that tell you about reading a schematic at face value?

Discussion and Conclusions

Summarize what you measured and what it confirmed. Address at least: whether the simplified circuits reproduced the originals on every row you tested, which law did the most work in each simplification, and why testing five of the thirty-two possible input combinations of circuit (a) is not the same as proving the two circuits are equal.
